Data Science & Machine Learning | 80 Hours | 7 Modules

Module	Hours	Module Title	Topics Covered	Hands-On Labs	Mini Project / Capstone	Tech Stack
1	5	Python for Data Science	NumPy arrays, vectorised operations Pandas — DataFrames, groupby, merge, pivot Matplotlib & Seaborn for EDA File I/O — CSV, JSON, Excel List comprehensions, lambda, map/filter Virtual environments & Jupyter setup	EDA on IPL / Zomato dataset Data cleaning pipeline lab Visualisation challenge (5 chart types) Pandas groupby analysis assignment	Mini Project: Exploratory Data Analysis on India's COVID-19 dataset — uncover state-wise trends, recovery rates, and vaccination progress with 10+ charts and a written summary report.	Python, NumPy, Pandas, Matplotlib, Seaborn, Jupyter
2	14	Statistics & Probability for ML	Descriptive statistics — mean, median, variance Probability distributions (normal, binomial, Poisson) Hypothesis testing — t-test, chi-square, ANOVA Correlation vs causation Central limit theorem & sampling Bayes' theorem basics	Hypothesis testing on sales data Distribution fitting with scipy A/B test simulation lab Correlation matrix heatmap project	Mini Project: A/B test analysis for an e-commerce website — compare two pricing strategies using hypothesis testing, calculate p-values, and recommend the better strategy with statistical co	Python, scipy, statsmodels, seaborn
3	16	Supervised Learning	Linear & logistic regression Decision trees & random forests Support vector machines (SVM) K-nearest neighbours (KNN) Gradient boosting — XGBoost, LightGBM Model evaluation — accuracy, F1, ROC-AUC Cross-validation & hyperparameter tuning	House price prediction with linear regression Titanic survival classification Feature importance with random forests GridSearchCV tuning lab	Mini Project: Loan default prediction system — build, compare, and evaluate 4 classifiers (LR, RF, XGBoost, SVM) on a banking dataset, select the best model, and explain results using featu	scikit-learn, XGBoost, LightGBM, GridSearchCV, Matplotlib
4	16	Unsupervised Learning & Feature Engine	K-means & hierarchical clustering DBSCAN for density-based clustering PCA & dimensionality reduction Feature scaling — StandardScaler, MinMax Encoding — OHE, label, target encoding Handling missing values & outliers Feature selection techniques	Customer segmentation with K-means PCA on image data (MNIST) Feature engineering on Kaggle dataset Outlier detection lab	Mini Project: Customer segmentation for a retail brand — use K-means clustering on RFM (Recency, Frequency, Monetary) features to identify 4-5 customer segments and build a targeting s	scikit-learn, KMeans, DBSCAN, PCA, Pandas

5	15	Deep Learning & Neural Networks	Perceptron, activation functions, backpropagation Building ANNs with Keras / TensorFlow Convolutional neural networks (CNN) Recurrent networks — LSTM, GRU Transfer learning — VGG16, ResNet Dropout, batch normalisation, regularisation Model saving, loading & deployment basics	MNIST handwritten digit classifier Image classification with CNN Sentiment analysis with LSTM Transfer learning fine-tuning lab	Mini Project: Plant disease detection system — train a CNN using transfer learning (ResNet) on leaf images to classify 10 disease types with 90%+ accuracy, wrapped in a simple Gradio web	TensorFlow, Keras, ResNet, VGG16, LSTM, Gradio
6	14	NLP & Time Series	Text preprocessing — tokenisation, stemming, TF-IDF Bag of words vs word embeddings (Word2Vec) Sentiment analysis & text classification Named entity recognition (NER) with spaCy Time series decomposition — trend, seasonality ARIMA, SARIMA forecasting models FB Prophet for business forecasting	Amazon review sentiment classifier Stock price ARIMA forecasting News topic modelling with LDA Prophet sales forecast lab	Mini Project: E-commerce review intelligence system — classify reviews by sentiment, extract key product complaints using NER, and forecast weekly sales trends using ARIMA, presented ir	spaCy, NLTK, Word2Vec, ARIMA, FB Prophet, scikit-learn
TOTAL — DS & ML		80 hrs				